Electrophysiology at the level of single neurons.

The neuron is the fundamental unit of the brain. To understand how the brain operates we must understand the operation of those single units. The neuronal spiking activity can be recorded by placing a high-impedance electrode close to the soma of the cell. Because this is invasive, it is normally only done in animals after rigorous ethical assessment. Sometimes this can be done in patients as part of their treatment. Whatever the context, there are a few procedures that are important and you will practice them today. These are:

- Recognize if the electrode is in active tissue or not.
- Detect action potentials in the signal. Sort spike according to their shape
- Decide if the activity comes from a single cell or multiple cells.

The data we will use today was recorded during surgery in Parkinson's disease patients. It is recorded and analyzed online during the surgery to identify the optimal target to place a Deep Brain Stimulation (DBS) probe.

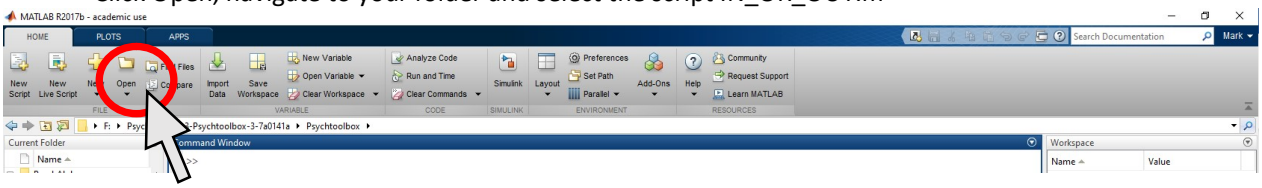
Part 1: Electrode placement

The first recordings are made ~10mm above the target. At that depth, the electrode tip is in white matter where there are no neurons nearby. The signal records mainly noise. As the electrode is lowered towards the target it eventually reaches the nucleus and the electrode starts to pick up neuronal activity (as well as noise). An experienced electrophysiologist can recognize if an electrode is in or out by the sight and sound of the signal.

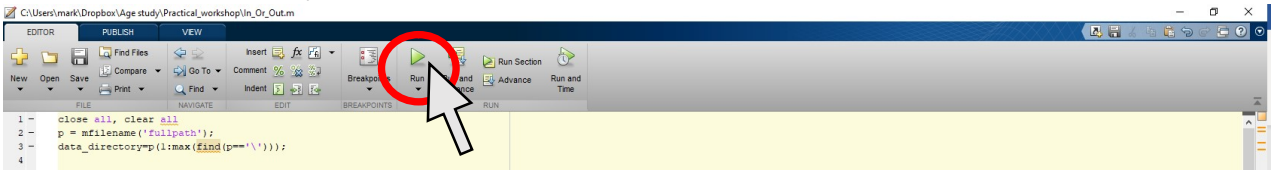
In the first exercise you will learn to tell if an electrode is in or out of active (neuronal) tissue

Open and Run IN_OR_OUT.m

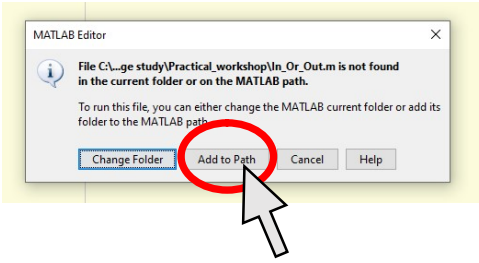
Click Open, navigate to your folder and select the script IN_OR_OUT.m



Click Run, at the top of the window



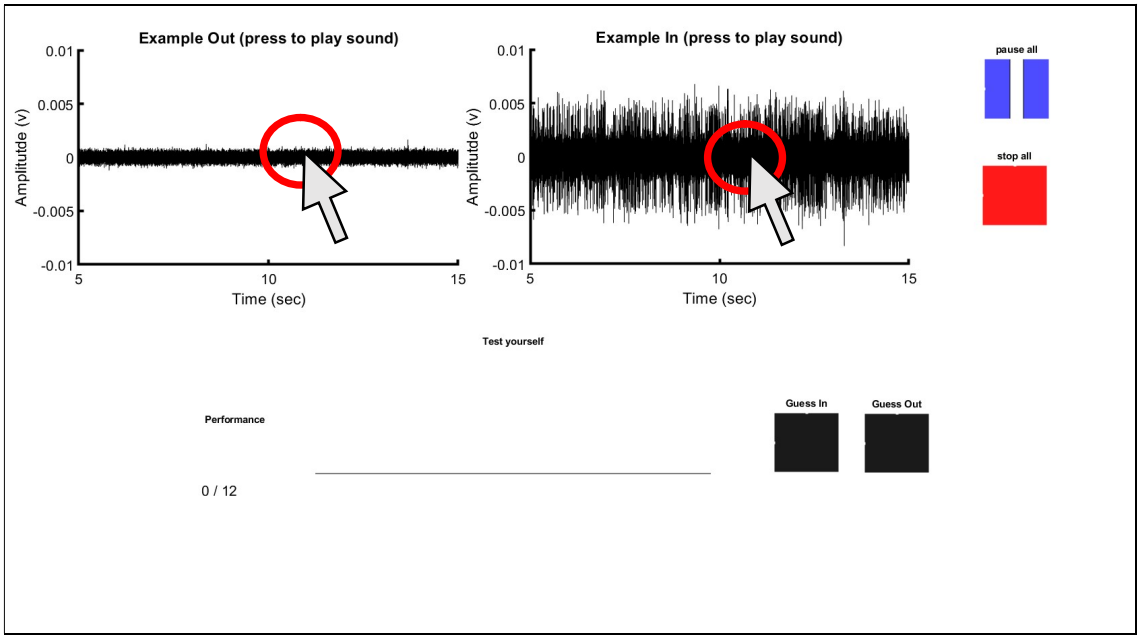
You will see this window, select Add to Path



The script now runs and displays the following figure.

Exercise 1.1

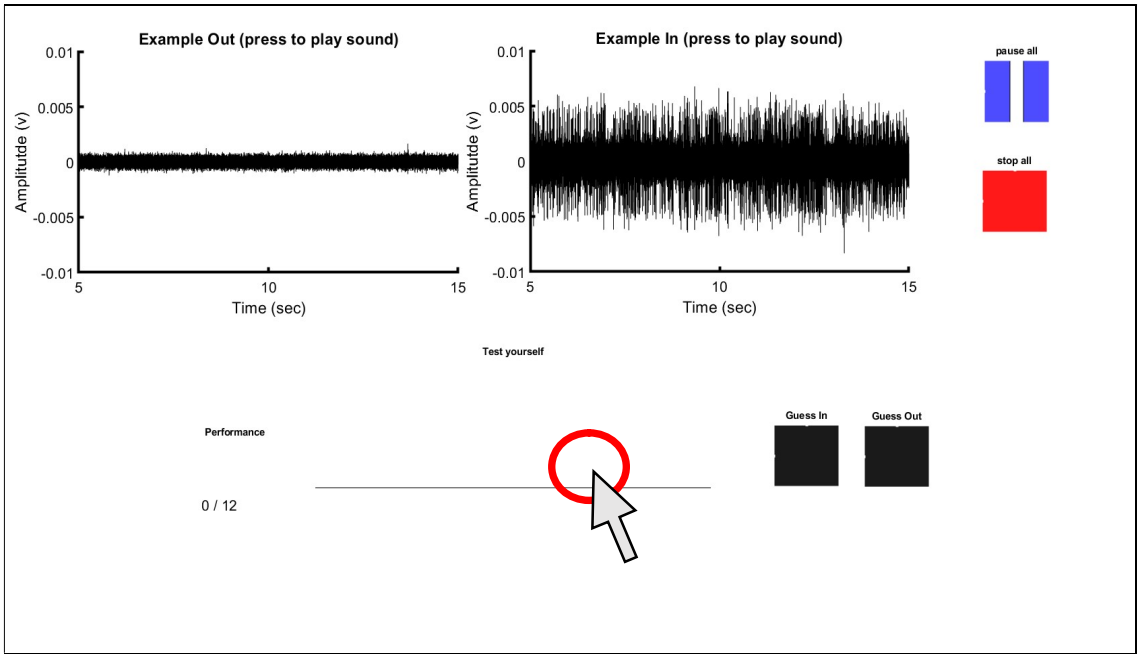
Click on the left and right upper plot to listen to the two example sound files. Do this a few times until you recognize the difference.



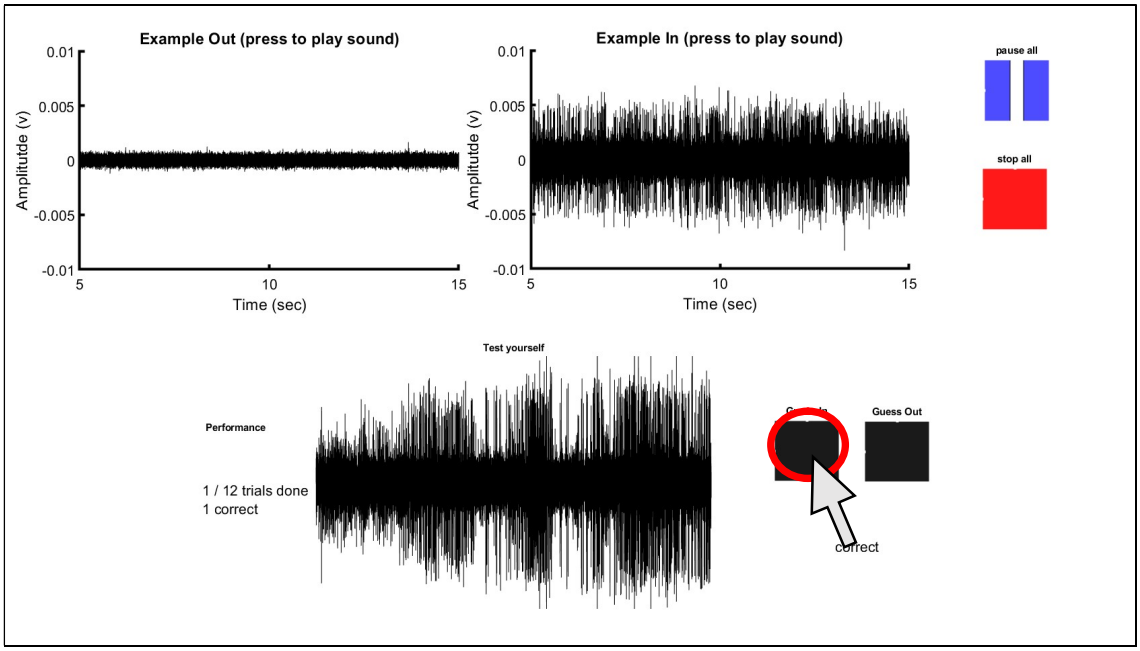
Exercise 1.2

When you can recognize the difference, try classifying 12 examples.

Click on the lower plot to play and show an example:



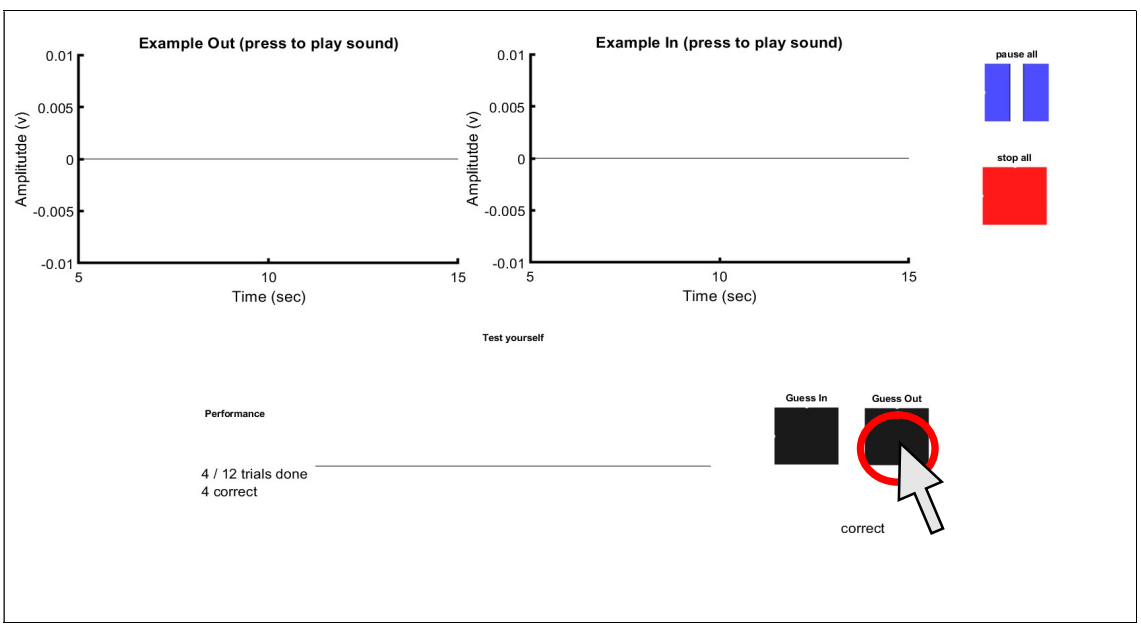
For each example click it you think the recording comes from in or out of the STN:



How did you do?

Exercise 1.3

Now try classifying recordings by sound alone. Open and Run IN_OR_OUT2.m. You will see that the exercise is very similar to the Exercise 1.2 but now you don't see the data plotted.



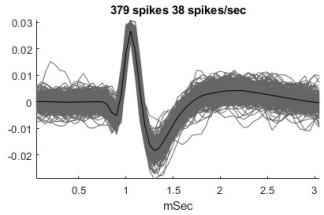
How did you do?

Part 2: Spike Detection

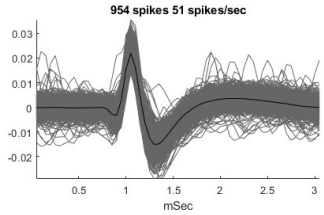
Action potentials can be discriminated from the rest of the signal (i.e. the noise) by their size.

There are many ways to detect and identify spikes in the noisy signal. The most basic and widely used way is a ***'window discriminator'***. This uses a lower threshold to detect spikes and an upper threshold to discard artifacts. Every time the signal goes into the window a spike is recorded.

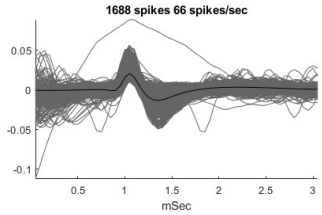
The art of it is to get the thresholds at just the right level; if you have the window too narrow, you miss spikes. If you have it too wide, you include noise. You may need to **cut** the data to a specific section if the cell is lost or to avoid nosy sections. You can tell if you have done a good job setting the window by inspecting the distribution of spikes shapes. There should be as many spikes as you can find without too much variance in the spike shape.



Too narrow window
missing spikes



Good window
More spikes, good shape



Too wide window
many spikes, not various shapes

Exercise 2.1

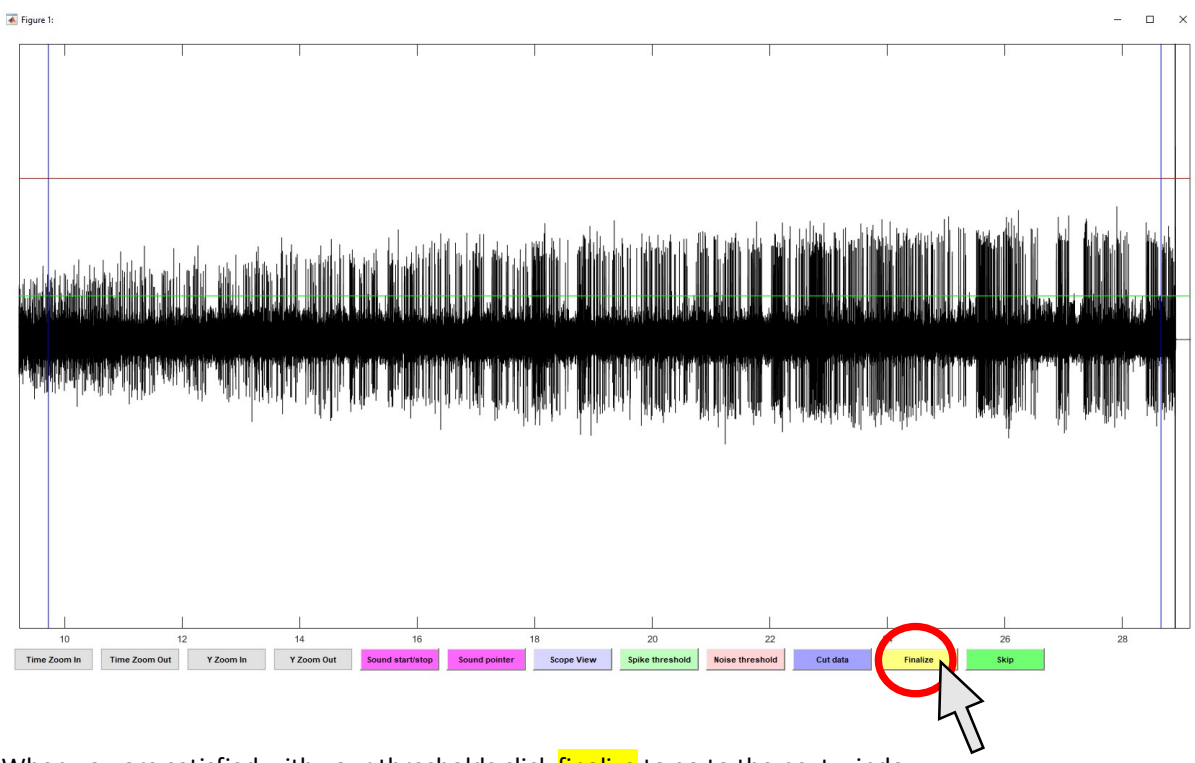
In the second exercise you will try using a window discriminator to detect spikes. Open and Run Spike_Detection_Exercise1.m

In the first stage you use the buttons along the bottom of the figure to set the spike threshold (lower threshold) and noise threshold (upper threshold). You can use cut data to set the start and end edges to exclude noisy periods.

1. For each threshold, clicking the button gives you crosshairs to set the threshold location.
2. Place the crosshairs and click with the left mouse button to set the threshold.
3. If you want to change the threshold click the button again.
4. For cutting the data you nee to place the crosshairs **twice**, at the start and end of the stable part of the recording

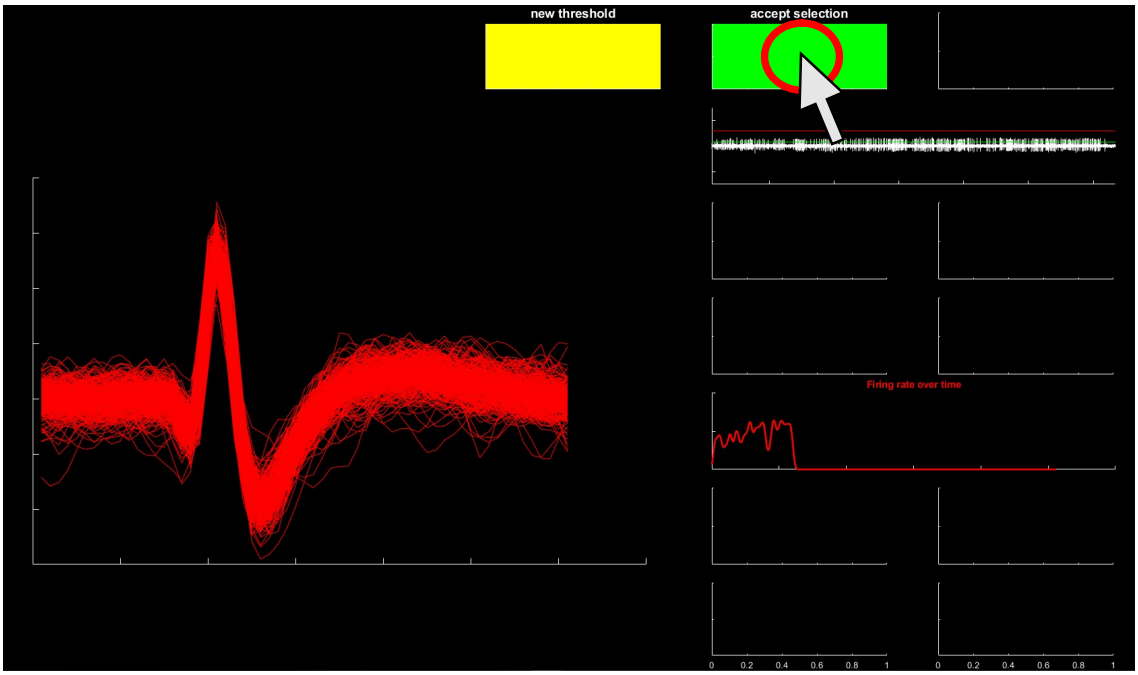
You can also use this window to inspect the data

1. You can zoom in the x axis to inspect and interesting part using time zoom in, then click an interesting point on the graph. Use Time Zoom out to reset the time axis.
2. Set the Y axis using Y zoom in and use the crosshairs to set the top and bottom of the y axis. Use Y Zoom out to reset the time axis.
3. You can play or stop the recording sound using Sound start/stop
4. You can select the start point for playing the sound using Sound pointer.
5. You can watch and hear the data in a rolling view, as it looks during the live recording using scope view.



When you are satisfied with your thresholds click finalize to go to the next window:

Hint: if you put the spike threshold too high (so that very few spikes are selected) the next window will not run.



- This window shows the spike waveforms that you have collected. If you are happy with the selection click accept selection otherwise click new threshold to re-do setting thresholds.
- After clicking accept selection you will move on to the next example.
- Set thresholds for 5 examples.

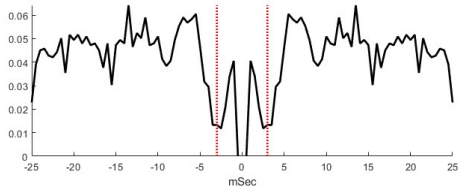
When you have finished the 5 examples you can see the results in the new figure. Your results are saved in a figure in the results folder.

Part 3: Single cell or many cells?

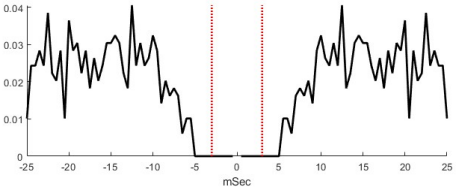
The electrode picks up action potentials from cells close to the tip. If there is more than once similar sized cell at an equal distance from the electrode tip, the electrode picks up activity from both. Therefore the question arises, is the activity coming from one cell (a 'single unit' recording) or from several (a 'multi-unit' recording).

We can tell the difference by looking at the minimum timing between any two spikes. Neurons have a *refractory period* of ~3ms after each action potential during which they cannot produce a second action potential. Therefore, if the electrode records two spikes within 3ms of each other, we can conclude that they come from different neurons.

The distribution of timing between spikes is quantified by the *autocorrelation* function. This is a histogram of how many spikes come at any given time offset from each other. The autocorrelation of a single unit has a characteristic gap of at least ±3ms in the center of the plot.



A multi-unit with many spikes in the refractory period



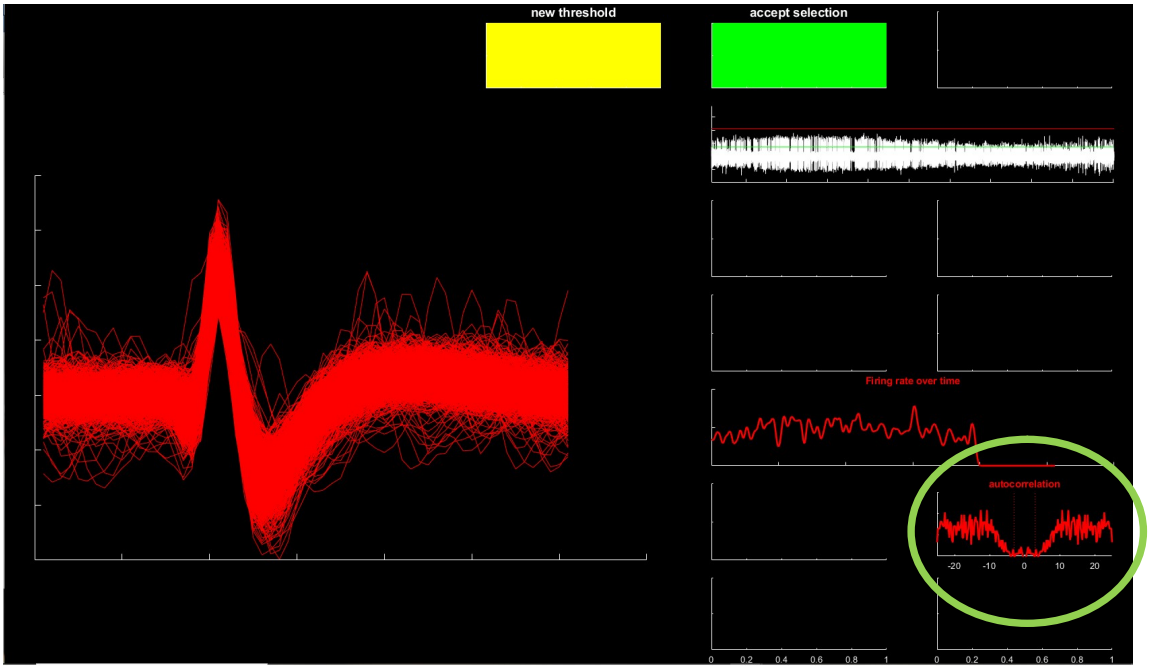
A single unit

Exercise 3.1

In the third exercise you will again use the window discriminator to detect spikes. Now also look at the autocorrelation to see if you are recording single units or multi units.

Open and Run Spike_Detection_Exercise2.m

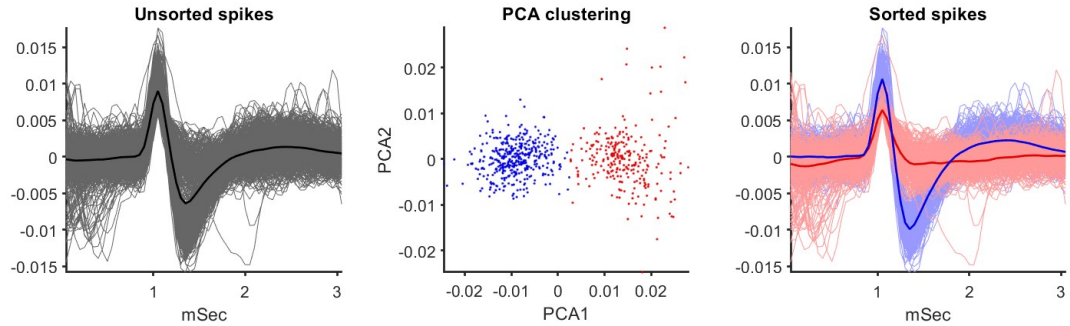
Set the thresholds as before. In the second window you will see a *new part of the figure*; the autocorrelation. The vertical dashed lines mark ±3ms, the autocorrelation should be low in this region.



Complete the 5 examples. You may need to adjust the thresholds until you get a characteristic single unit autocorrelation. In some recordings, you will not manage this! Your results are saved in a figure in the results folder.

Part 4: Sorting based on the spike shape.

The shape of the action potential is informative about the placement of the electrode and the type of neuron. Because electrodes tend to pick up several neurons (and noise), we can use the spike shape to separate the spikes from different neurons. But which features of the spike should we use to classify them? Principal Components Analysis (PCA) looks for the most informative features and clustering algorithms cluster spikes together into likely groups. We can then inspect the autocorrelation to see whether any spike clusters represent single neurons.



Thresholds produce unsatisfactory results

PCA reveals two clusters

Sorting based on PCA reveals two spike waveforms

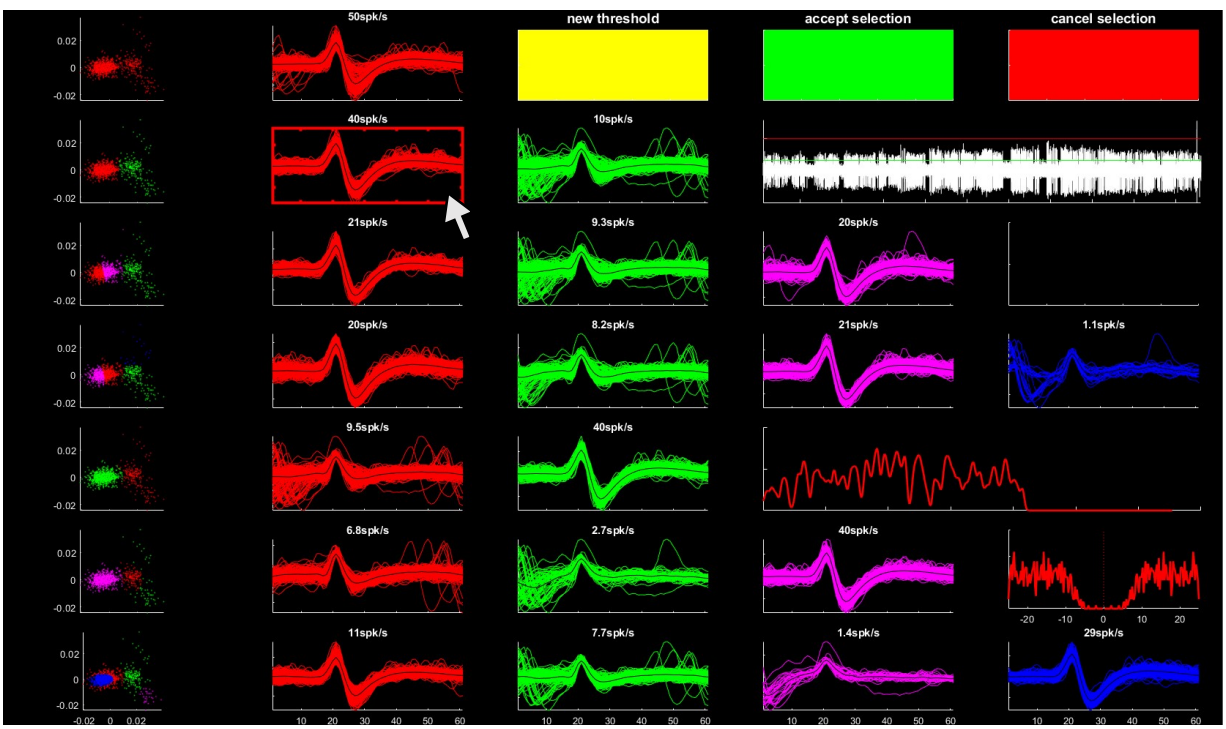
Exercise 4.1

In the fourth exercise you will again use the window discriminator to detect spikes, you will then further sort them using PCA and clustering.

Open and Run Spike_Detection_Exercise3.m

Set the thresholds as before. The second window shows the results from PCA and clustering. The first column shows each spike plotted in PCA space, colored according to clusters. The spike waveforms of each cluster are plotted along each row.

Select a good cluster by clicking on the spike waveform plot. If you are happy with the selection click accept selection to go to the next example. Otherwise cancel selection to choose a different cluster or new threshold to redo the thresholding.



Complete the 5 examples. Your results are saved in a figure in the results folder.

Part 5: Put everything together

Now combine everything you have learned. Run Spike_Detection_Exercise4.m

for each example

- Decide if the electrode is in or out of the STN (if you think it is not in the STN click finalize without setting a spike threshold, or click accept selection without choosing a cluster)
- For electrodes that have good spikes set the window discriminator.
- Select good clusters from the PCA space.
- Inspect the autocorrelation and decide if you have single units or multiunits.
- Complete the 10 examples.
- Your results are saved in a figure in the results folder.